Apex PdM: Hear It Before It Breaks

A low-cost sensor pod that helps township teams fix small machine problems before they become big breakdowns.

Submission Details

- **Track:** PLANET, 03 Energy Efficiency
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- **Department:** [Enter each member's department]
- **Date of submission:** [Insert date] (deadline Friday 23 October 2026, 11:59 PM)
- **Live demo:** https://washer-pdm.pages.dev
- **Prototype / code:** [Insert repository link]
- **Stage:** Working prototype (TRL 4-5) with live data online

1. Problem Understanding and Overview

Mosques, suraus, kuils, malls and clubhouses in a township run a lot of spinning machines like air-conditioning compressors, water pumps and fans. Most of the time they only get fixed after they break. So we built a working sensor rig to change that.

1.1 The Problem

What is the problem with machine maintenance today? Maintenance is mostly reactive. A US Department of Energy guide found that over 55% of an average facility's maintenance is still "run it till it breaks" (US DOE FEMP, *O&M Best Practices Guide*, Release 2.0, 2004). Note these are US figures and not Malaysian or township data. Before a machine fails a worn bearing, damper or seal slowly makes it shake more, sound different and work harder. Nobody notices until it stops.

What makes it hard to spot wear early? Industrial vibration sensors cost thousands of ringgit each so only big factories use them. Technicians check machines one by one by hand and still cannot see the wear happening inside. Real faults are also rare so there is very little fault data to train a detector on.

Who does it affect? A small part worth tens of ringgit can turn into a new motor or compressor worth thousands plus days without cooling. When an AC unit fails the first people to find out are the ones sitting under it, often during Friday prayers, Ramadan nights or weddings. A worn machine also uses more electricity and gets thrown away sooner.

1.2 Business Goals

- Help township facility teams move from fixing breakdowns to catching early signs of wear. This way they can plan a cheap repair instead of a full replacement and keep shared spaces running.
- Fit the *Township of Tomorrow* PLANET theme through energy efficiency (Track 03, which covers HVAC optimisation). Air-conditioning is the largest energy user in Malaysian office buildings, about 57% (Saidur, 2009) [2]. Faults that come from equipment wear raise building energy use; in the US, typical building faults are estimated to cost 103–500 TWh of extra energy a year (Melgaard et al., 2022, citing Roth et al., 2004) [3]. Our smart plug measures each machine's power directly, so a unit that starts working harder shows up before it breaks. Fixing a small part instead of replacing the whole machine also supports the circular economy: machines last longer and less scrap goes to landfill.

1.3 Objectives

1. Build a pod that mounts on the outside of a machine and records vibration, sound and power together.

2. Learn what "normal" looks like for each machine without needing examples of broken machines.
3. Show live data on a web dashboard that anyone can open.
4. Prove it on a hard test machine (a front-load washer) before moving to AC compressors and water pumps.
5. Keep the kit cheap enough to fit on every unit (about RM 170 per pilot kit today).

2. Proposed Solution

Apex PdM is a small sensor pod that sits on the outside of a machine. It feels the shaking, hears the grinding and measures the power. Then it sends the data to the internet and shows it live on a website. A machine learning model learns what normal looks like and raises an anomaly score when a run looks different.

2.1 Solution Overview

How it flows

Sensors → ESP32 → Wi-Fi / Internet → Cloud database → Live website (the machine learning model sits on the data and sends an alert)

A dual-core ESP32 runs the pod. One core reads the sensors in real time while the other handles Wi-Fi and cloud upload. Every 5.024 seconds the pod sends a 45-column record and 32 of those columns are signal features (audio 15, vibration 13, power 4). If the Wi-Fi drops it keeps about 4 minutes of data and sends it once the connection is back.

AI and machine learning This is unsupervised anomaly detection and not generative AI. The model is trained only on normal runs so it learns what "healthy" looks like for that machine and needs no broken-machine examples. It uses feature fusion by combining vibration features (how much the machine shakes and its main frequency), sound features (loudness in different pitch bands) and power use. We compared three detectors: Mahalanobis distance, Isolation Forest and One-Class SVM. To test them fairly we left out one whole wash cycle at a time.

Each machine learns its own normal Every machine gets its own baseline instead of one generic threshold. Next we plan to save the machine state (wash, rinse or spin from Samsung SmartThings) with the data and compare inside the same phase. That way a normal fast spin is not mistaken for a fault. At township scale identical units could also be compared with their neighbours so the odd one out stands out (proposed).

Business process improvement Technicians are told which machine needs a look instead of rushing to emergency calls. Repairs can be planned ahead and done outside busy times.

2.2 Step-by-Step Approach

Step 1: Fit the pod on the outside of the machine. The pod sits in an ABS project box (158×90×60 mm, splash-resistant but untested) stuck to the outer chassis with 3M VHB tape. Nothing inside the machine is opened. A Tapo P110 smart plug measures power without any rewiring. Photos of the rig are in Appendix A.

Step 2: Turn raw signals into numbers on the chip. The MPU-6050 samples vibration at 200 Hz (±16 g) and the INMP441 microphone samples sound at 16 kHz. Firmware 07 turns these into features on the chip and uploads only the numbers so no sound recordings ever leave the device. The dashboard polls the cloud database no faster than every 30 seconds.

Step 3: Learn normal and flag anomalies. Our first test used leave-one-wash-out on 8 earlier healthy washes. These were recorded in an older snapshot format (one 5 s snapshot each) and not the 32-feature firmware-07 windows. The result was 0 of 8 washes falsely flagged (95% upper bound 31%) and a simulated doubling of sound was caught in 7 of 8. Added vibration was not detected in that test and our report puts this down to the older data format. The firmware-07 model needs at least 3 cycles and 2 were recorded as of 20 Oct.

Step 4: Move to township machines. We will record 20+ more cycles to measure the false-alarm rate. Then we test on an air compressor and a water pump and after that run a pilot on about ten AC units at one township site. The longer-term plan is a custom board, an industrial accelerometer and one dashboard for the whole township.

2.3 Data and Input Sources

Where the data comes from The sensor pod (vibration and sound) and the Tapo P110 smart plug (power) are mounted on a Samsung WW10TP44DSX front-load washer. We chose a washer because it is easy to access and also a hard test: it shakes hard, the load moves, water sloshes and the speed keeps changing. An air compressor and a water pump run right beside it so there is real background noise too.

What kind of data Numeric features for each 5-second window: vibration statistics and main frequency, sound loudness and pitch bands, and power. Machine state (wash, rinse, spin) is not saved yet but it is planned. As of 20 Oct we have 11 wash sessions recorded and 8,379 training-ready rows including 2 complete firmware-07 cycles.

How it is processed Features are computed on the ESP32. Python pipelines standardise the rows into one training dataset. The model is trained and tested one cycle out at a time so it is never tested on data it has already seen.

How fresh it is A new window arrives every 5.024 seconds and power is read every second. The dashboard updates live during a cycle and the baseline is retrained as new healthy cycles are recorded.

3. Key Features of the Solution

Feature 1: Three sensors working together One sensor alone gets fooled because a normal fast spin can look like a fault. Checking vibration, sound and power together gives a fuller picture. On the washer the vibration jumped over 20× from idle to spin and the main frequency climbed to 19.9 Hz (about 1,200 rpm) in both firmware-07 washes. Washer sound in the higher bands was 36-90× louder than room noise even with the compressor and pump running nearby.

Feature 2: Private by design Sound is turned into numbers on the chip and never uploaded as a recording. This matters in homes and places of worship and it keeps data and cloud costs low.

Feature 3: Learns each machine's normal with no fault data needed Real faults are rare so a model that needs only healthy runs can start on a new machine straight away.

Feature 4: Sees energy waste, low cost and easy to fit The sensor node cost RM 131.07 in parts and the full pilot kit cost RM 169.68 with the smart plug (net paid, from receipts; see Appendix B). It mounts on the outside and plugs in with no rewiring. Because it measures power directly it can compare each run's power use with that machine's own normal and show when the machine starts working harder.

4. Expected Outcomes and Benefits

4.1 Business Benefits

Resident wellbeing and satisfaction Repairs can be planned ahead so cooling stays on during Friday prayers, festivals and busy weekends.

Financial and environmental value Energy (our estimate, to be confirmed in the pilot): one 5 kW commercial AC running 10 hours a day uses about 18,250 kWh a year. If a worn part makes it use 10% more, that is about 1,825 kWh or **about RM 900 a year** at roughly RM 0.50/kWh. The 10% is an assumption, not a measurement. It sits inside the 5–20% energy-bill savings the US DOE guide estimates for O&M programmes that target energy efficiency [1, §2.4]. A pod costing about RM 170 that catches one such problem pays for itself within a few months. Reliability: surveys cited by the same US DOE guide found predictive maintenance eliminated 70–75% of breakdowns, cut maintenance costs by 25–30% and reduced downtime by 35–45% [1, §5.4] (US figures from 2004 and not our own result). Less waste: fixing a small part instead of replacing a whole unit also cuts scrap.

Maintenance cost and scalability Pods that are cheap enough for every unit make it practical to watch all the AC units in a mosque or mall from one dashboard. Our target users are facility managers of mosques, suraus, kuils, malls and clubhouses, local maintenance contractors and the residents who use these spaces.

4.2 Risk Management

Risk	How we handle it
Early evidence only: one test machine, 8 evaluated cycles and simulated (not real) faults. Added vibration was not detected in the first test.	Record 20+ more cycles on the new firmware and train the firmware-07 model. We need about 30 alarm-free cycles to claim a false-alarm rate under 10%. Real fault runs come after that.
Moving from a washer to AC units and pumps is not proven yet.	Test on the compressor and pump at home first then run a pilot on about ten AC units.
Background noise from nearby machines.	Take noise baselines with the nearby machines on and off.
The box has drilled ports so we make no IP rating claim.	Weatherproof the box before any outdoor use.
Code credentials may have been shared by mistake.	Rotate the secrets and check the repo history before sharing the code link.
Cloud usage limits.	Keep dashboard polling at 30 seconds or

Risk	How we handle it
	slower and upload only features.

5. Conclusion

Addressing the problem Township machines get fixed after they break. Apex PdM gives them a low-cost pod that feels, hears and measures each machine, learns its normal and flags early signs of wear. A small repair then replaces a big breakdown.

Why this solution works It is a real working rig streaming live data from a hard test machine today. It combines three sensors for about RM 170, needs no fault data and uploads no recordings. It also has a clear path from one pilot site to one dashboard for the whole township. Our ask is one township site for a pilot.

6. References

1. US Department of Energy, Federal Energy Management Program. *Operations & Maintenance Best Practices: A Guide to Achieving Operational Efficiency*, Release 2.0 (PNNL-14788, July 2004), §2.4, §5.2 and §5.4. https://www.pnnl.gov/main/publications/external/technical_reports/pnnl-14788.pdf (US figures from 2004. The reactive share comes from a 2000 study.)
2. Saidur, R. (2009). Energy consumption, energy savings, and emission analysis in Malaysian office buildings. Energy Policy, 37(10), 4104–4113. https://doi.org/10.1016/j.enpol.2009.04.052 (air conditioners are the major energy users, 57%, in Malaysian office buildings).
3. Melgaard, S. P., Andersen, K. H., Marszal-Pomianowska, A., Jensen, R. L., & Heiselberg, P. K. (2022). Fault Detection and Diagnosis Encyclopedia for Building Systems: A Systematic Review. Energies, 15(12), 4366. https://doi.org/10.3390/en15124366 (open access; it cites Roth, K. W., Westphalen, D., Llana, P., & Feng, M. (2004), The Energy Impact of Faults in U.S. Commercial Buildings, International Refrigeration and Air Conditioning Conference, Purdue, https://docs.lib.purdue.edu/iracc/665).

Appendix A: The Prototype Rig

These photos show the working Apex PdM rig that streams the data on the live dashboard (https://washer-pdm.pages.dev). All photos were taken by the team in October 2026.



Figure A1. The sealed pod mounted on the outside of the Samsung WW10TP44DSX test washer. Nothing inside the machine is opened.

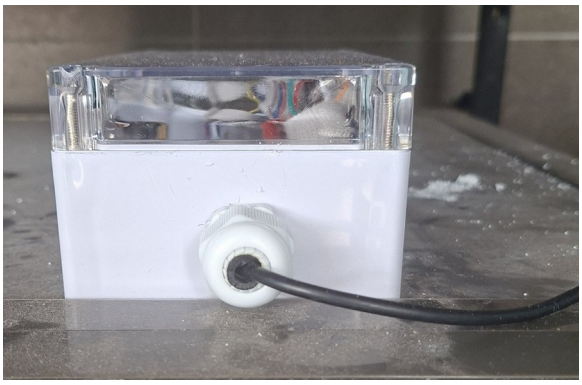


Figure A2. The sealed pod: ABS box (158×90×60 mm) with a cable gland for the USB power lead, stuck on with 3M VHB tape.



Figure A3. Inside the pod: the ESP32 on its screw-terminal expansion board, with the sensor wiring tied down against vibration.

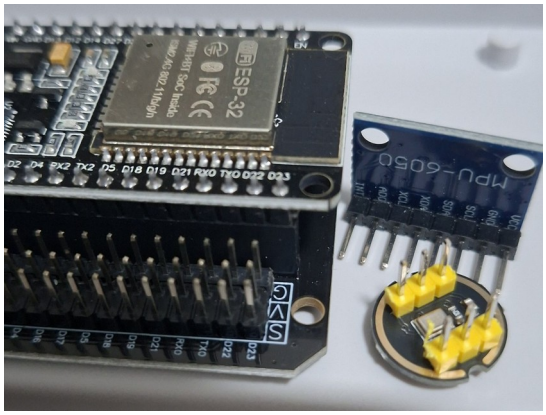


Figure A4. The sensing parts: ESP32 dev board (left), MPU-6050 vibration sensor (top right) and INMP441 microphone (bottom right).



Figure A5. The off-the-shelf parts as delivered, before assembly: enclosure, ESP32, expansion board, MPU-6050, INMP441, screws and jumper wires. The Tapo P110 smart plug (power sensor) is not pictured.

Appendix B: Hardware Cost (Shopee Malaysia Receipts, October 2026)

Every part was bought off the shelf. The totals below are the amounts actually paid per order, after vouchers and discounts.

Order	Parts	Store	Paid (RM)
1	NodeMCU ESP32 dev board (ESP-WROOM-32, CP2102, 30-pin), with a free micro-USB cable	Robotedu.my Malaysia Robotics	32.90
2	ESP32 30-pin screw-terminal expansion board (RM 8.46) + 40 Dupont jumper wires, M-F 20 cm (RM 3.20)	MakerHub MY	10.47
3	MPU-6050 6-axis vibration sensor (RM 21.99) + INMP441 I2S digital microphone (RM 27.99), pre-soldered	ElectricA Ecom Malaysia	49.98
4	ABS junction box enclosure, 158×90×60 mm	baba168.my	16.30
5	PG7/PG9/PG11 nylon cable glands + 50 insulated ferrules	CASSETECH Electrical Store	5.55
6	3M VHB heavy-duty mounting tape, 25 mm × 5 m (list RM 37.60)	3M shop (Shopee Mall)	15.87
7	TP-Link Tapo P110 smart plug with energy monitoring (list RM 45.00)	Tapo Malaysia (Shopee Mall)	38.61
	Total, complete pilot kit (7 orders)		**169.68**

The sensor node alone (orders 1–6) cost RM 131.07; the Tapo P110 smart plug adds RM 38.61.

Source: Shopee Malaysia order receipts, October 2026, as compiled in the team's hardware cost breakdown. Order totals are the amounts paid after vouchers and discounts; list prices are shown in brackets where an order held two parts or had a discount.